Basic Network Security Assessment

Project 2 • Cybersecurity Portfolio Lab

Organization: SmallTech Solutions (fictional laboratory environment)

Role: Cybersecurity Analyst

Platform: Kali Linux virtual machine in VirtualBox

1. Executive Summary

This assessment was conducted as a controlled cybersecurity laboratory exercise for SmallTech Solutions, a fictional small-office environment. A Kali Linux virtual machine was assessed to identify network services, review selected security configurations, assess potential risks, and provide recommendations.

Testing included network discovery, TCP and UDP port scanning, service/version identification, SSH configuration review, firewall inspection, and basic connectivity testing. SSH was identified on TCP port 22 with password authentication enabled. The inspected INPUT firewall chain had an ACCEPT default policy with no individual INPUT rules listed.

These conditions may increase the potential attack surface if SSH becomes reachable by an unauthorized party. The work was performed in a controlled VirtualBox laboratory environment; it does not establish public-internet exposure. Positive controls included restricted password-based root login and disabled empty-password authentication.

2. Scope and Methodology

Scope: The primary target was the Kali Linux virtual machine running in VirtualBox. The assessment focused on network exposure, services, SSH configuration, firewall configuration, and basic connectivity.

Method: Discover → Identify → Analyze → Assess Risk → Recommend → Report.

Tools: Nmap, ip, ping, ss, systemctl, sshd, and iptables.

Testing was limited to the controlled laboratory environment. No unauthorized external systems were tested and no exploitation was performed.

3. Technical Findings

3.1 Network Configuration and Connectivity

eth0 used IPv4 10.0.2.15/24 with default gateway 10.0.2.2. Ping to the gateway returned 0% packet loss, confirming connectivity. The neighbor table showed 10.0.2.2 as REACHABLE.

3.2 TCP Port and Service Discovery

Nmap identified 22/tcp as open and associated with SSH. A full TCP scan found only port 22 open. Service/version detection identified OpenSSH 9.3p2 Debian 1. The version was documented; no specific vulnerability was attributed solely from version identification.

3.3 UDP Port Assessment

The top 20 UDP ports tested were all closed. This was not a complete UDP port-range scan.

3.4 SSH Configuration

SSH was active and running. Effective settings included passwordauthentication yes, permitrootlogin without-password, permitemptypasswords no, and maxauthtries 6. Password authentication is a potential area for improvement because reachable password-based SSH can be exposed to password-guessing attacks. Root password-based SSH login was restricted and empty passwords were prohibited.

3.5 Firewall Configuration

The inspected iptables INPUT chain had an ACCEPT default policy and no individual INPUT rules. An iptables-nft management notice was also observed. This does not establish that the host has no firewall or that it is publicly reachable.

3.6 SSH Network Accessibility

Nmap against 10.0.2.15 confirmed 22/tcp open. This demonstrates that SSH is reachable through the VM's network interface, but does not prove reachability from another physical system or the public internet.

4. Risk Assessment and Recommendations

Finding	Likelihood	Impact	Risk
SSH password authentication enabled	Medium	High	Medium–High
No restrictive INPUT rules observed	Medium	Medium	Medium

Finding 1 — SSH password authentication: Prefer SSH key-based authentication where appropriate; disable password authentication only after key-based access has been securely configured and tested; restrict SSH access to trusted sources where practical; monitor authentication logs.

Finding 2 — INPUT firewall policy: Use a restrictive/default-deny approach for unnecessary incoming traffic, explicitly permit required services, restrict SSH access to trusted sources where possible, and review firewall rules regularly.

Positive controls: Root password-based SSH login is restricted, empty-password authentication is disabled, authentication attempts are limited, and only one open TCP service was identified in the full TCP scan.

5. Conclusion

SSH on TCP port 22 was the primary network service identified. SSH was active and reachable through the VM's network address. Password-based SSH authentication was enabled, representing an area for improvement. The inspected INPUT firewall chain also used an ACCEPT default policy with no individual rules observed.

The assessment also identified useful controls, including restrictions on password-based root login and prohibition of empty passwords. Because the work was performed in a controlled VirtualBox environment, the findings should not be interpreted as evidence of public exposure or compromise.

Stronger SSH authentication, restricted unnecessary incoming traffic, and regular review of services and firewall configuration would strengthen the assessed system.

6. Evidence Appendix

Key commands:

```
ip -br addr
ip route
ping -c 4 10.0.2.2
nmap 127.0.0.1
nmap -sV 127.0.0.1
nmap -p- 127.0.0.1
sudo nmap -sU --top-ports 20 127.0.0.1
systemctl status ssh --no-pager
sudo sshd -T
sudo ss -tulnp
sudo iptables -S
sudo iptables -L INPUT -n -v --line-numbers
nmap -p 22 10.0.2.15
ip neigh
```

Key evidence: 10.0.2.15/24; gateway 10.0.2.2; 0% packet loss; 22/tcp open; OpenSSH 9.3p2 Debian 1; SSH active/running; password authentication enabled; root password-based SSH login restricted; empty passwords disabled; MaxAuthTries 6; INPUT policy ACCEPT; no individual INPUT rules listed; 22/tcp open on 10.0.2.15; gateway REACHABLE.